Innovation, Business Models and Entrepreneurship

Week 5: Assignment 5

Q1 Which of the following is correct?

(A) Engineering->Technology->Science

(B) Science->Engineering->Technology

(C) Science->Technology->Engineering

(D) Technology->Engineering->Science

Q2 Statement I: All scientific inventions lead to technological innovation.

Statement II: Technological innovation is when an invention is transformed into a socially usable product.

Which of the following is correct?

(A) Only Statement I is correct.

(B) Only Statement II is correct.

(C) Both the statements are correct.

(D) Both the statements are incorrect.

Q3 Which of the following is not true regarding technological innovation?

(A) It is about creating new use of the existing technology.

(B) It is a complex process involving technological, social and economic factors.

(C) It requires huge financial investments.

(D) It takes long period of time, sometimes decades for such innovations to be impactful.

Q4 Which of the following are associated with technological innovation process:

(a) Basic research

(b) Product Development

(c) Societal impact- positive and negative

(d) Co-creation

(e) Excluding existing knowledge

(f) Scaling up

(g) Trial and error

(h) Experimentation

Please select the correct option:

(A) (a), (b), (d), (f), (h)

(B) (a), (b), (c), (d), (f)

(C) (a), (b), (e), (f), (g)

(D) (b), (d), (e), (f), (h)

Q5 Technological innovation can be divided into following catagories:

Types-

(a) Radical innovation

(b) Micro-radical innovation

(c) Generational innovation

(d) Incremental innovation

Examples-

(i) Virtual trial rooms

(ii)Smartphone camera enhancement

(iii) Creation of world wide web

(iv) Transition from 3G to 4G mobile network

Please select the correct match below:

(A) a-ii, b-i, c-iii, d-iv

(B) a-i, b-iv, c-ii, d-iii

(C) a-iii, b-i, c-iv, d-ii

(D) a-iv, b-iii, c-i, d-ii

Q6 Which of the following is not true:

(A) S-curve helps stakeholders anticipate the trajectory of a technology's development and plan for future developments and challenges.

(B) Incremental innovation involves short term planning while micro-radicle and generational innovation involves long term planning.

(C) Radical innovations are rare phenomenon and are difficult to plan based on S-curve.

(D) India majorly has 3rd generation companies where R&D planning is embedded in corporate and business planning.

Q7 Based on competitive impact, technologies can be divided as:

(a) Base technologies

(b) Key technologies

(c) Pacing technologies

(d) Emerging technologies

Match with the correct statements-

(i) Technologies well embodied in products and are basis of competitive advantage as they provide differentiation to companies.

(ii) Technologies which already demonstrated their ability to change the basis of competition and as a result are now moved to the development stage.

(iii) Ageing technology which is available or known to all industry participants but cannot provide competitive advantage because of its widespread use.

(iv) Forward looking technologies at an early research stage with unknown potential, but promising and could provide competitive advantage in future.

Please select the correct option below:

(A) a-i, b-ii, c-iii, d-iv

(B) a-iii, b-i, c-ii, d-iv

(C) a-iii, b-i, c-iv, d-ii

(D) a-i, b-iii, c-ii, d-iv

Q8 Which of the following statements are true for strategy?

(A) Offensive strategies help maintain the status quo and usually help expand the existing markets.

(B) Absorbent strategies involve acquiring license to exploit innovation in domestic market.

(C) Dependent strategies focus on developing core strategy with the parent company.

(D) Imitative strategies are research intensive strategies.

Q9 Which of the following are technology forecasting tools?

(a) Time series analysis

(b) Morphological analysis

(c) Delphi technique

(d) Moving average

(e) Pearl function based on S-curve

(f) Envelop curve and trend extrapolation

Please select the correct option:

(A) (a), (d), (e), (f)

(B) (a), (b), (c), (e)

(C) (b), (c), (e), (f)

(D) (b), (d), (e), (f)

Q10 Statement I: Technology forecasting is important for businesses to anticipate any future technological developments, trends, innovations and to develop strategies to take advantage of future opportunities and mitigate any potential threat to the existing business.

Statement II: These forecasting tools assist organizations in making informed decisions about research, development, investment, and strategic planning.

Which of the following is correct?

(A) Only Statement I is correct.

(B) Only Statement II is correct.

(C) Both the statements are correct.

(D) Both the statements are incorrect.